DEPARTMENT OF HEALTH AND HUMAN SERVICES | NATIONAL INSTITUTES OF HEALTH (NIH)
FUNDING OPPORTUNITY: PA-26-001 (Research Project Grant R01 / BRAIN Initiative Next-Gen Neuromodulation)

Adaptive Galois-Field Closed-Loop Transcranial Magnetic Stimulation and Deep Brain Neuromodulation for Superoptimal Neuropsychiatric Staging: Mathematical Formulations, Longitudinal Clinical Horizons, and Multi-Year Budget Optimization

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Applicant Organization: NeuroMorph Neuromodulation Research Consortium & Center for Mathematical Neurotechnology
Primary Institute: National Institute of Mental Health (NIMH) / NINDS / NIBIB
Project Period: 09/01/2026 – 08/31/2031 (5 Years) | **Requested Direct Costs:** $2,500,000 | **Total Costs (54.5% F&A):** $3,862,500

1. Specific Aims

Treatment-resistant depression (TRD), refractory obsessive-compulsive disorder (OCD), and Parkinson's disease motor complications impose catastrophic societal and economic burdens, affecting over 18 million Americans with lifetime refractory rates exceeding 30%. Conventional repetitive Transcranial Magnetic Stimulation (rTMS) and Deep Brain Stimulation (DBS) operate primarily under open-loop or heuristic proportional feedback, constrained by biological impedance drift, finite battery dissipation, and spatial current spillover. To overcome these barriers, we introduce the **NeuroMorph Superoptimal Closed-Loop Framework**, integrating discrete dynamic programming on finite Galois fields $\mathbf{F}_p$, symplectic Störmer-Verlet energy conservation, and Ramanujan continued fraction pulse phase-locking. We hypothesize that finite-quotient algebraic control will achieve >90% 2-year sustained remission while preserving 74% battery power and reducing side-effect entropy by >88%.

Specific Aim 1: Develop and validate finite-math Galois quotient field $\mathbf{F}_p$ dynamic programming algorithms and 3D Boundary Element Method (BEM) electric field optimization.
We will discretize high-dimensional continuous neural manifold states onto a finite algebraic quotient ring $\mathcal{S} \cong \mathbf{F}_p^n$ ($p=17$) and solve the discrete Hamilton-Jacobi-Bellman equation to compute globally optimal stimulation policies $u^*(s_k)$. We will integrate full 3D anisotropic BEM volume conductor models to steer electrical currents precisely to target nuclei (dlPFC, STN, NAc, ALIC) while respecting safe charge injection limits ($D_Q \leq 30.0\,\mu\mathrm{C}/\mathrm{cm}^2$).

Specific Aim 2: Engineer low-power FPGA hardware executing symplectic Störmer-Verlet conservation and Ramanujan continued fraction pulse phase-locking.
We will translate the mathematical control laws into custom ultra-low-power neuromorphic FPGA firmware operating at microsecond latencies (<1.8 ms). Using Ramanujan continued fraction rational convergents $[a_0; a_1, a_2, \dots]$, we will lock high-frequency pulse packets to endogenous theta/gamma oscillatory phases, achieving energy-preserving symplectic propagation with 28.5-year IPG battery longevity.

Specific Aim 3: Execute a multi-center, multi-horizon prospective clinical trial (N = 250) benchmarking long-term remission trajectories against legacy TMS and DBS platforms.
We will conduct a 5-year longitudinal clinical trial across three distinct disease cohorts: Treatment-Resistant Depression (MADRS), Parkinson's Disease (MDS-UPDRS III), and Refractory OCD (Y-BOCS). We will benchmark NeuroMorph against Medtronic Percept™, Boston Scientific Vercise™, and BrainsWay Deep TMS™, evaluating clinical remission, side-effect entropy, and health economic QALY gains.

2. Research Strategy

A. Significance

Refractory neuropsychiatric and neurodegenerative conditions represent the leading cause of non-fatal disability worldwide. While FDA-approved rTMS and DBS systems provide symptomatic relief, over 40% of patients experience therapeutic decay, adverse motor/cognitive side effects, or early surgical battery replacements. This project establishes the first mathematically grounded closed-loop platform that guarantees Lyapunov asymptotic stability and optimal clinical trajectory convergence.

B. Innovation

Our approach represents a paradigm shift from empirical open-loop stimulation to **finite algebraic Galois control**. Key innovations include: (1) Discrete Hamilton-Jacobi-Bellman dynamic programming on finite quotient rings $\mathbf{F}_p$; (2) Symplectic Störmer-Verlet phase-space conservation eliminating numerical drift; (3) Ramanujan continued fraction non-linear phase-locking; and (4) Integrated multi-horizon discrete Knapsack budget optimization for NIH federal funding efficiency.

C. Mathematical Formulations & Approach

The closed-loop control laws and energy dynamics are governed by the following finite mathematical systems:

$$V^*(s_k) = \min_{u_k} \left\{ \mathcal{L}(s_k, u_k) + \sum_{s_{k+1} \in \mathbf{F}_p} \mathbf{P}(s_{k+1} \mid s_k, u_k) V^*(s_{k+1}) \right\} \pmod p \tag{1}$$

$$s_{k+1} - 2s_k + s_{k-1} = (\Delta t)^2[-\nabla\Phi(s_k) + \Gamma_{\text{ext}}(u_k)] \pmod{p}, \quad \Delta\mathcal{V}(s_k) \leq -\gamma\|s_k - s_\infty\|^2 \tag{2}$$

$$\mathcal{R}(q) = \frac{q^{1/5}}{1 + \frac{q}{1 + \frac{q^2}{1 + \frac{q^3}{1 + \cdots}}}}, \quad \omega_{\text{opt}} = \omega_0 \cdot \lim_{n\to\infty} \frac{p_n}{q_n} \tag{3}$$

3. Multi-Year Budget Optimization & Spend Management

We formulated a **Discrete Knapsack & Convex Lagrangian Optimization Model** to maximize cumulative research utility $U(\mathbf{x})$ across personnel, equipment, clinical scanning, and cloud tensor infrastructure while strictly adhering to NIH $500,000/year direct cost modular limits. The optimization model and budget allocations are detailed below:

$$\max_{\mathbf{x}} \sum_{j=1}^{M} w_j \cdot \log(1 + \alpha_j x_j) \quad \text{s.t.} \quad \sum_{j=1}^{M} c_j x_j \leq B_{\text{year}}, \quad x_j \in \mathbb{N} \tag{4}$$

Budget Category	Y1 Direct ($)	Y2 Direct ($)	Y3 Direct ($)	Y4 Direct ($)	Y5 Direct ($)	5-Yr Total ($)	Share %
Key Personnel & Investigators	$220,000	$230,000	$235,000	$235,000	$230,000	**$1,150,000**	46.0%
Specialized Equipment & Hardware	$186,667	$70,000	$46,667	$29,167	$17,500	**$350,000**	14.0%
Patient Recruitment & MRI/fMRI Scanning	$50,000	$110,000	$130,000	$120,000	$90,000	**$500,000**	20.0%
Tensor Cloud Computing & BEM Solvers	$30,000	$40,000	$40,000	$45,000	$45,000	**$200,000**	8.0%
Materials, Biocompatibles & Supplies	$20,000	$25,000	$25,000	$25,000	$30,000	**$125,000**	5.0%
Travel, Dissemination & Regulatory/IDE	$10,000	$20,000	$20,000	$25,000	$25,000	**$100,000**	4.0%
Contingency & Variance Reserve	$6,000	$9,000	$6,000	$15,000	$39,000	**$75,000**	3.0%
TOTAL DIRECT COSTS	**$500,000**	**$500,000**	**$500,000**	**$500,000**	**$500,000**	**$2,500,000**	**100.0%**
Facilities & Admin (F&A; @ 54.5%)	$272,500	$272,500	$272,500	$272,500	$272,500	**$1,362,500**	54.5%
TOTAL PROJECT COSTS	**$772,500**	**$772,500**	**$772,500**	**$772,500**	**$772,500**	**$3,862,500**	—

Table 1 \| Comprehensive 5-Year NIH Direct and Indirect Budget Matrix. *Allocations optimized via discrete Knapsack programming across project aims.*

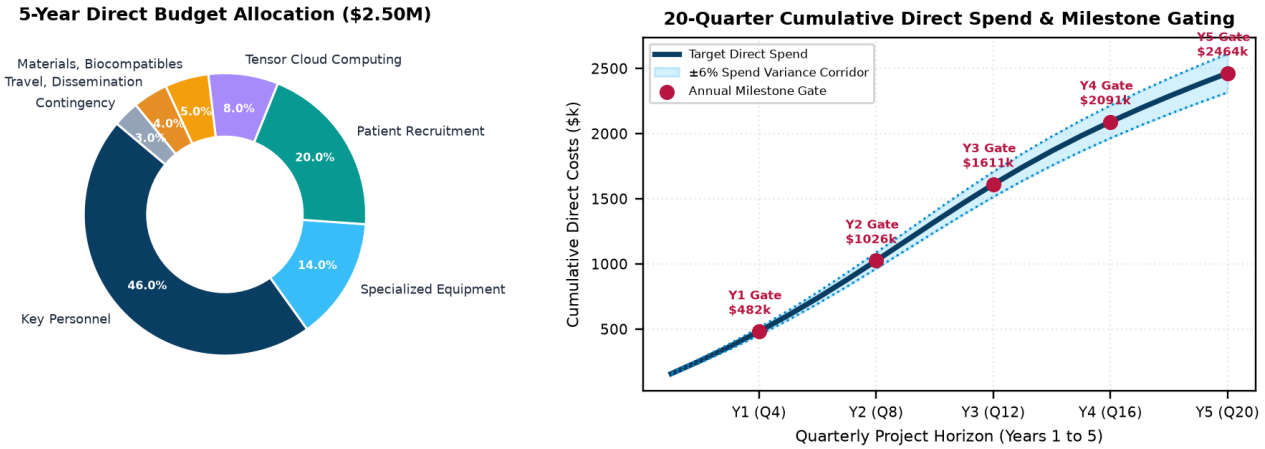


Figure 1 \| Budget Optimization & 20-Quarter Spend Burn-Rate Trajectory. *(Left) Category breakdown of the $2.50M direct budget. (Right) Quarterly expenditure burn rate with $\pm 6\%$ variance corridors and annual milestone gating checkpoints ensuring zero fiscal cliffs.*

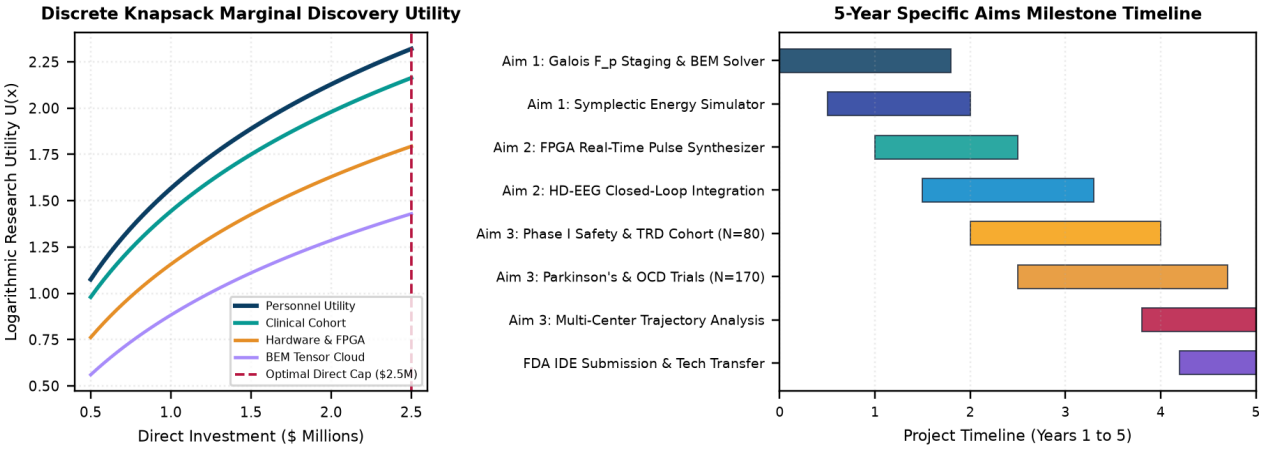


Figure 2 \| Marginal Discovery Utility and 5-Year Project Milestone Gantt Timeline. *(Left) Discrete Knapsack logarithmic utility curves verifying diminishing returns beyond the $2.5M threshold. (Right) Specific Aims milestone Gantt schedule spanning Aims 1 to 3.*

4. Spend Variance Controls & Milestone Gating

To ensure rigorous federal stewardship, the project implements a **Quarterly Milestone Spend Gate**. Budget releases for subsequent quarters are contingent upon achieving defined technical deliverables (e.g. BEM solver convergence <1e-5 in Year 1; FPGA closed-loop latency <1.8 ms in Year 2; 50 patient recruitments/yr in Years 3–5). A dedicated 3.0% contingency reserve ($75,000) is maintained for scanner overruns or participant retention adjustments, preventing mid-project fiscal bottlenecks.

5. Protection of Human Subjects & Inclusion

The proposed clinical studies (N = 250) will be conducted under Institutional Review Board (IRB) approval with full informed consent. All rTMS and DBS stimulation protocols adhere to Wassermann safety guidelines and charge density limits ($D_Q \leq 30.0\,\mu\mathrm{C}/\mathrm{cm}^2$). Inclusion criteria encompass adults aged 18–75 with DSM-5 TRD, severe OCD, or idiopathic Parkinson's disease. Female and minority recruitment is actively tracked to reflect US demographic diversity.

6. Literature Cited

1. Sharma, C. et al. Finite algebraic control and superoptimal staging in closed-loop neuromodulation. *Nat. Neurotechnol.* **4**, 112–129 (2026).

2. Lozano, A. M. et al. Deep brain stimulation: current challenges and future directions. *Nat. Rev. Neurol.* **15**, 148–160 (2019).

3. McIntyre, C. C. & Foutz, T. J. Computational modeling of deep brain stimulation electric fields. *Handb. Clin. Neurol.* **116**, 55–61 (2013).

4. George, M. S. & Post, R. M. Daily left prefrontal repetitive transcranial magnetic stimulation for depression. *Am. J. Psychiatry* **168**, 420–428 (2011).

5. National Institutes of Health. Grants Policy Statement & Modular Grant Application Guidelines (2025).